

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

Paper 3

9648/03

16 September 2014

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [] at the end of each question or part question.

Section A

Answer all questions.

Write your answers in this booklet using dark blue or black pen.

Section B

Write your answers in this booklet using dark blue or black pen.
For Free Response Question, write your answers on the separate Answer Paper using dark blue or black pen.

For Examiner's Use	
1	/ 7
2	/ 8
3	/ 11
4	/ 14
5	/ 12
6	/ 20
TOTAL	/ 72

This document consists of **14** printed pages.

[Turn over

Section A

Answer all questions.

- 1 Restriction fragment length polymorphism, RFLP, refers to differences in fragment length generated by restriction enzymes. These differences are used to detect variations in genomes.

Fig. 1.1 below shows the inheritance of a common human trait which is linked to a restriction fragment length polymorphism (RFLP). Genomic DNA from each family member was assayed for a linked restriction fragment length polymorphism (RFLP). EcoRI sites on Southern blots using probes derived from the gene was shown.

Use the pedigree from the following family along with the Southern blot analyzing inheritance of the RFLP to answer the following questions.

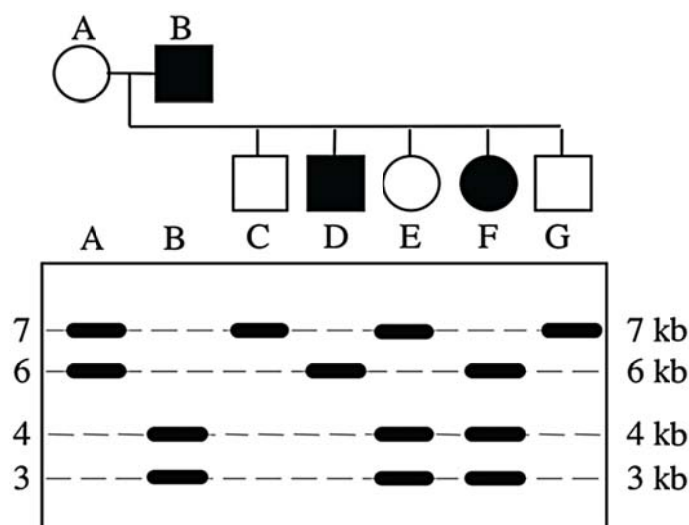


Fig. 1.1

- (a) What is the mode of inheritance for this trait?

[1]

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- (b) Sketch a suitable diagram to illustrate the region of genomic DNA recognised by the probe in individual A and F. Show the positions of EcoRI sites and the distance between them in kilobases (kb).

[2]

- (c) A different study was carried out on another family. Fig. 1.2 below shows a Southern blot with an RFLP detected by using a probe for a sequence located on a sex chromosome. The variant exhibits two forms, one at 7 kb and one at 6 kb. The mother, father and daughter are normal.

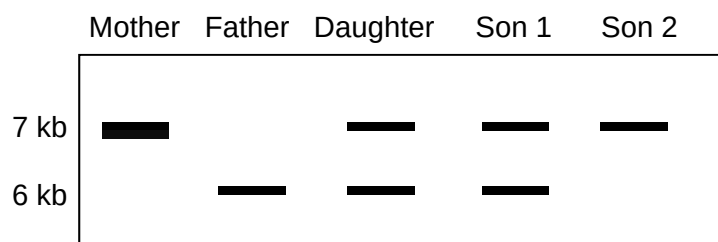


Fig. 1.2

- (i) Which son is abnormal?

[1]

- (ii) Explain how you arrived at your answer.

[1]

Variable number tandem repeats (VNTRs) are repetitive, non-coding sections of DNA. A particular VNTR is located at the same locus in different individuals, but the number of repeats in that VNTR varies between individuals.

- (d) Explain how, in the process of genetic fingerprinting, gel electrophoresis is able to distinguish between the VNTRs that occur at the same loci of different individuals.

[2]

[Total: 7 m]

- 2 The artificial plasmid, pBR322, was constructed to act as a vector. It has often been used to insert human genes, such as the human insulin gene, into the bacterium, *Escherichia coli*.

The plasmid was constructed to include two genes, each giving resistance to a different antibiotic: an ampicillin resistance gene and a tetracycline resistance gene. The plasmid also has a target site for the restriction enzyme, BamHI, in the middle of the tetracycline resistance gene.

A pBR322 plasmid was cut using BamHI and the cDNA gene for human insulin inserted into it. Fig. 2.1 shows pBR322 and the recombinant plasmid.

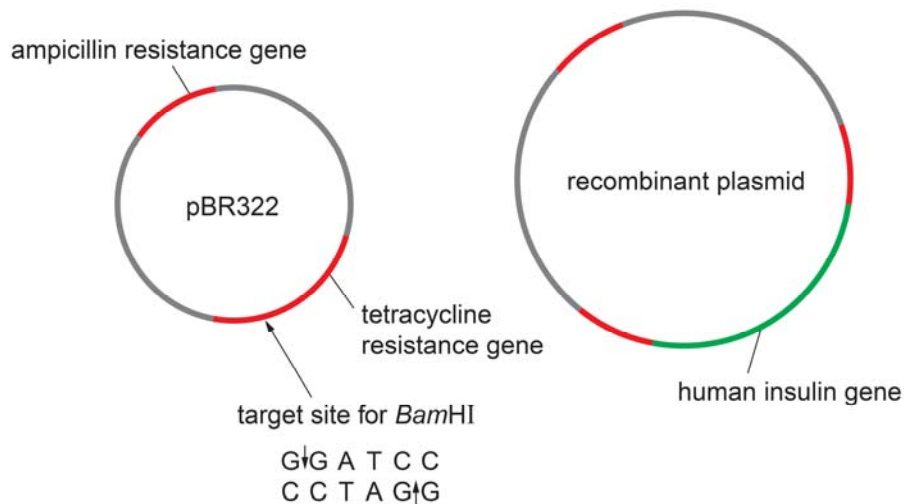


Fig. 2.1

- (a) With reference to Fig. 2.1, describe how a cDNA human insulin gene can be inserted into pBR322 that has been cut by BamHI.

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[3]

- (b) Bacteria were then mixed with the recombinant plasmids. Those bacteria which had successfully taken up recombinant plasmids were identified using the following steps:

step 1 – the bacteria were spread onto culture plates containing nutrient agar and ampicillin and incubated to allow colonies to form

step 2 – some bacteria from each of the colonies growing on these plates were transferred to plates containing nutrient agar and tetracycline, as shown in Fig. 2.2.

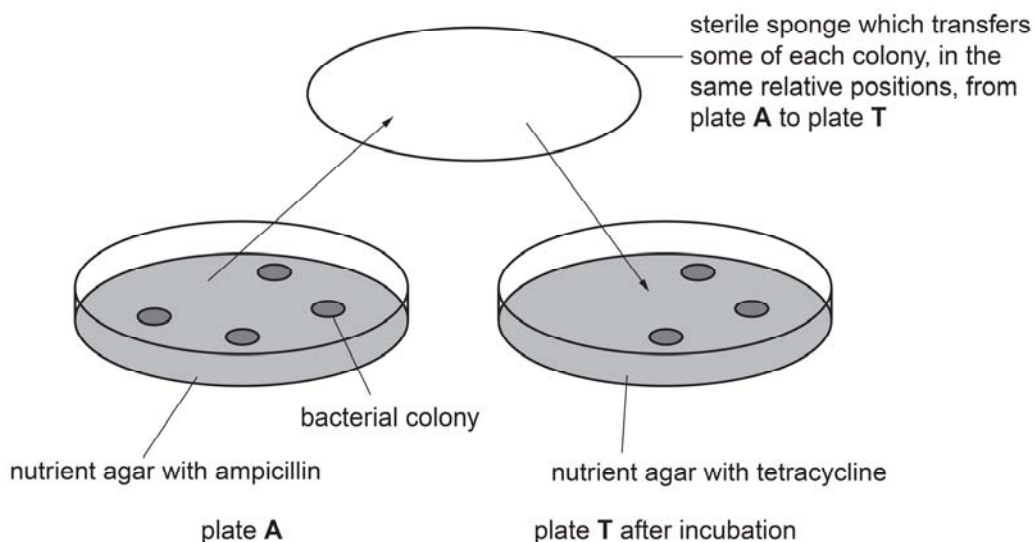


Fig. 2.2

- (i) Explain why the bacteria were first spread onto plates containing ampicillin.

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[1]

- (ii) With reference to Fig. 2.1, explain why it is important for *Bam*HI target site to be within tetracycline resistance gene.

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[2]

(iii) Use a labelled line and the letter **C** to identify, on Fig. 2.2, a colony of bacteria that contain the recombinant plasmid.

[1]

(iv) Plasmid vectors carrying antibiotic resistance genes are now rarely used in gene technology. Explain why antibiotic resistance genes are now rarely used.

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[1]

[Total: 8 m]

3 (a) One of the treatments for cystic fibrosis (CF) is to inhale a mist of sterile hypertonic saline (an intensely salty solution). Explain how hypertonic saline can be an effective treatment for CF.

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[2]

(b) Gene therapy involves transferring normal CFTR alleles into lung epithelial cells with defective genes. Describe how a liposome can be used to insert the normal CFTR allele into a CF patient.

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[2]

(c) "Gene therapy for CF does more than treating the symptoms of the genetic disease; it targets the cause of the disease."

Explain why this statement is true.

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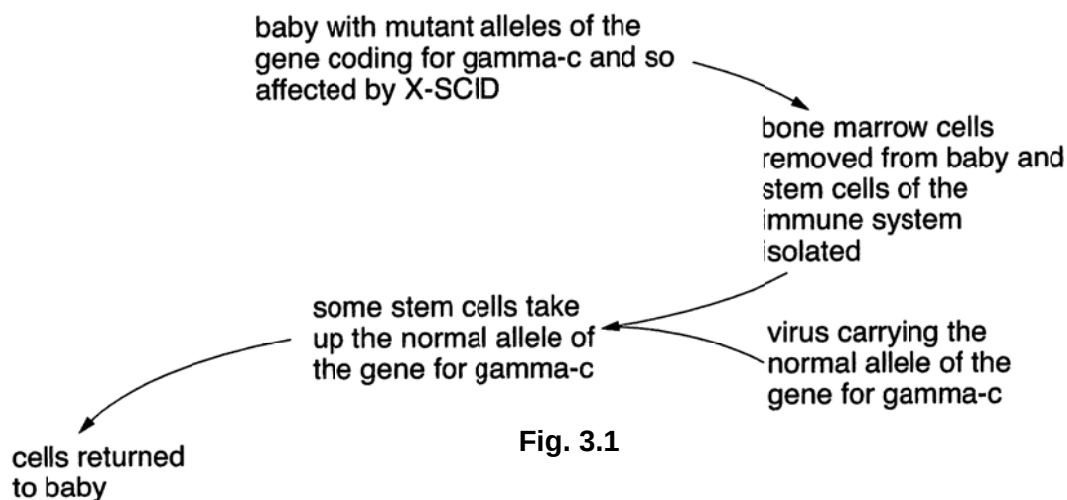
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[1]

- (d) Explain why it is easier to perform gene therapy on recessively inherited diseases.

[3]

Gene therapy for several types of SCIDs depends on the isolation of stem cells of the immune system. X chromosome-linked SCID (X-SCID) is caused by an inability to produce a protein, gamma-c. The procedure for the gene therapy is shown in Fig. 3.1.



- (e) With reference to the procedure shown in Fig. 3.1,
 (i) suggest the type and potency of stem cells used in this treatment.

[1]

- (ii) describe the properties of the stem cells that are used.

[2]

[Total Marks: 11]

Name: _____ Class: 2bi2____ / 2IPbi2____

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Section B

Answer **all** the questions in this section.

- 4 Transgenic plants over-expressing the vacuolar H^+ -pyrophosphatase are much more resistant to high concentrations of NaCl and to water deprivation than the wild-type strains. These transgenic plants accumulate more Na^+ and K^+ in their leaf tissue than the wild type. Vacuolar H^+ -pyrophosphatase of Arabidopsis is encoded by a single gene, *AVP1*. In an experiment, agrobacterium-mediated transformation was performed on Arabidopsis thaliana to produce eight transgenic lines, with enhanced salt tolerance. The procedure of culturing cells already transfected with *AVP1* gene can be divided into four stages.

- (a) (i) Briefly describe the main idea of Stage 2 and 3.

Stage 1: Establishment of aseptic culture.

Explant is disinfected and grown in suitable medium containing nutrient and agar.

Stage 2: Multiplication and axillary shoot enhancement

..... [1]

Stage 3: Rooting

..... [1]

Stage 4: Acclimatisation

Plantlets transferred to soil in sheltered environment. Young plants are eventually replanted in an open uncontrolled environment.

- (ii) Identify one issue that can occur in stage 4.

..... [1]

- (b) Two of the cell lines (AVP1-1 and AVP1-2) were chosen for further analysis, along with the wild type.

The three cell lines were placed under a water-deficit condition for eight days. And results of their relative water content of wild type and AVP1 transgenic plants were recorded in Fig. 4.1

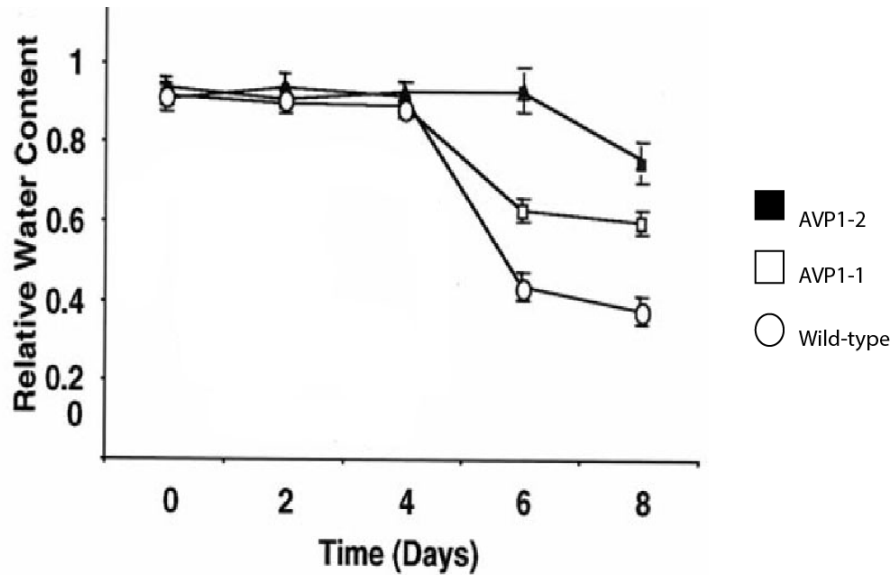


Fig. 4.1

- (i) With reference to Fig. 4.1, explain the effect of the transgene on the plantlets and drought-resistance.

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[4]

- (ii) Although drought-resistant plants can provide benefits to farmers in arid regions, they may also have detrimental effects to the environment. Suggest one such effect.

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[1]

- (c) Further research into *AVP1-1* and *AVP1-2* cell line showed that there is a distinct genetic difference between them, and both are tagged with a different genetic marker; *AVP1-1* with genetic marker 1 and *AVP1-2* with genetic marker 2.

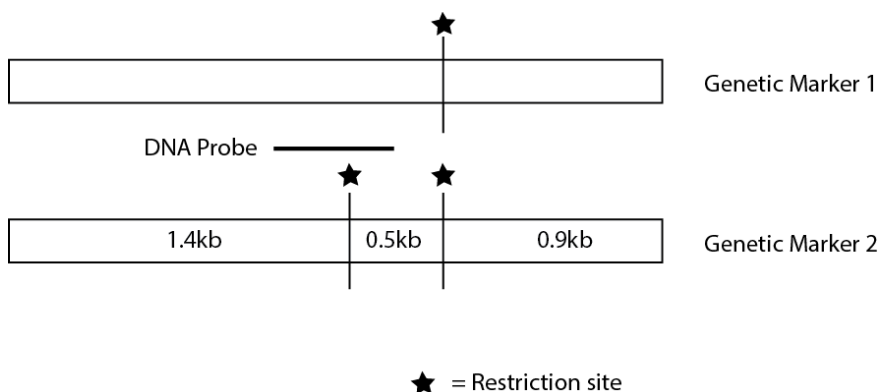


Fig. 3.2

- (i) Explain how restriction fragment length polymorphism comes about.

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[2]

- (ii) With reference to Fig. 3.2, describe how RFLP can be used to identify cell line carrying *AVP1-2* gene.

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[4]

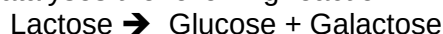
[Total: 14 m]

5 Planning Question

Write your answers to this question on the separate answer paper provided.

Inhibitors are substances that reduce the activity of enzymes. When the inhibitor binds reversibly to the active site of the enzyme, it is known as a competitive inhibitor. Often a competitive inhibitor is a similar shape to the substrate. When a non-competitive inhibitor combines with an enzyme, the shape of the active site is altered sufficiently to prevent the substrate from combining with it.

The enzyme β -galactosidase catalyses the following reaction:



The chemical ONPG (o-nitrophenyl β -D-galactopyranoside) is also degraded by the enzyme, β -galactosidase:



The ONP produced is yellow, allowing the rate of this reaction to be followed colorimetrically. The concentration of ONP is proportionate to its absorbance. Using this information and your own knowledge, design an experiment to determine whether iodine solution, an inhibitor of this reaction, is a competitive or non-competitive inhibitor.

You must use:

- 6 cm³ ONPG stock solution
- β -galactosidase stock solution
- 40 cm³ buffer pH 8
- 5 cm³ I₂/KI solution
- Colorimeter (440 nm filter)

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- cuvettes
- beaker of crushed ice
- droppers
- eye protection
- gloves

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- Include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 12]

Free Response Question

Write your answers to this question on the separate answer paper provided.

Begin each section of the question on a new piece of answer paper.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b), etc., as indicated in the question.

- 6 (a) Explain how insulin genes are cloned using *E. coli* cells to produce eukaryotic proteins. [6]
- (b) Compare and contrast between treatment of cystic fibrosis and SCID. [6]
- (c) Explain the factors that prevent gene therapy from becoming an effective treatment for genetic diseases. [8]

--- End of Paper ---